

[← Back to projects](#)[Healthcare](#)[Education](#)[Accessibility](#)

RAISEYOURIQ CASE STUDY

Turning cognitive science into a product children want to use.

A scientifically proven method packaged in a fun, accessible app that helps children with disabilities improve cognitive abilities—backed by 50+ years of research and achieving 90%+ success rate.

Role	Timeline
Product Designer	4 months
Team	Market
10 people	USA

Deliverables

[Mobile App](#)[Design System](#)[User Research](#)[Accessibility](#)[Gamification](#)

50+

years of cognitive science

90%+

reported success rate

AAA

accessibility ambition

STORY

• Overview

- Research
- Architecture
- Motivation
- Accessibility
- Outcome

THE ASSIGNMENT

Package 50 years of science into an app that feels safe, playful, and credible.

A unique algorithm developed over 50 years by scientists—a proven method to improve cognitive abilities. But it needed to reach children with special needs, individuals on the autism spectrum, and those seeking prophylactic care for Alzheimer's disease.

The mission: create an app where accessibility isn't just a feature—it's the foundation.



Scientific product, not another kids app

The only brain training system proven to increase intellectual abilities through peer-reviewed scientific studies.



Two audiences had to succeed

Parents needed evidence and control. Children needed a playful experience that did not feel like clinical work.



Accessibility was the product strategy

The app needed to serve children with special needs, users on the autism spectrum, and adults seeking prophylactic care for Alzheimer's disease.

RESEARCH

I couldn't design what I didn't understand.

Through rigorous analysis of scientific materials, I uncovered critical insights that shaped every pixel of the interface.

How I designed for people with disabilities

RESEARCH DEEP DIVE

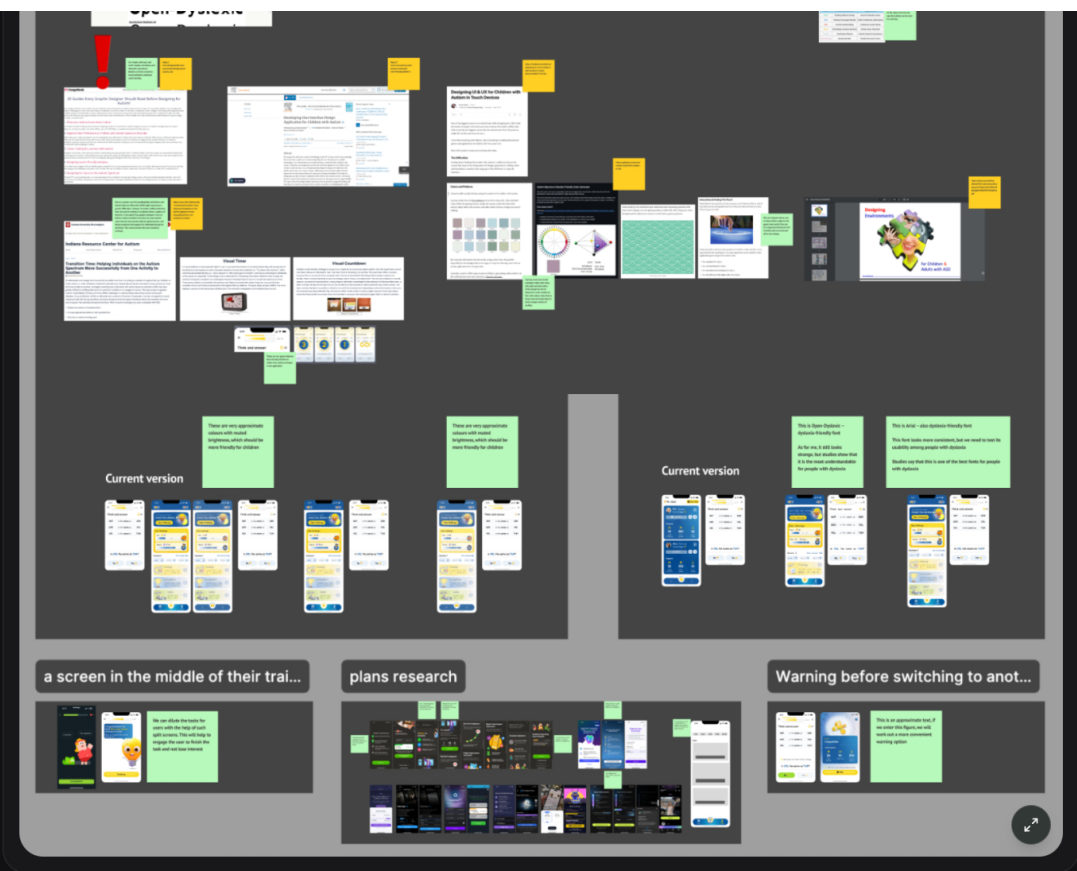
How I designed for those I couldn't understand

In this article, I go deeper into the autism research, accessibility decisions, and the insights that shaped RIQ's colors, typography, sensory controls, and time-based interaction patterns.

[Read the research article](#)

Research





Colors and Patterns

Intensive color usually distracts and gives anxiety to the children with autism.

A group study done by [G.A. Antshel](#) states that using soft, calm and dark colors while designing interior designs for rooms created for kids with autism, helps kids to focus more and allow them to have a longer session of studying.

By using the information by this study, using colors from this palette especially for the background of our apps is a key for allowing user to focus on the application for a longer time.

I actually created a little open-source tool that is generating colors relative to this informations and color structure. [Check it out here.](#)

Autism Spectrum Disorder Friendly Color Generator

Kids with autism can have an especially hard time managing their concentration while interacting with their environment. Some colors and patterns might also give them anxiety.

This tool is intended to generate colors for your User Interfaces that will be used by kids with autism. However, the colors that this tool generated can be used in things like book covers, page and typography design or something as simple as what color to paint a room.

How does it work?

- Subdued and colours mixed with grey were favored by the children with autism
- A predominant preference for colors in the Blue/Green hue sectors was notable
- A balance between colourfulness and greyness is seen to be popular

Studies on Color Preferences and Aversions

Research has shown that [children with autism often prefer certain hues](#), particularly those that are softer and less intense. Duller tones, pastels, and muted shades are generally more comforting to them. These colors can have a calming effect, helping to reduce feelings of anxiety or overstimulation.

Color Preferences and Effects

Color	Common Reaction in AS	Suggested Use
Green	Soothing, Reduces Anxiety	Good for Relaxation Areas
Blue	Calming, Encourages Serenity	Useful in Bedrooms, Quiet Spaces
Red	Can Be Overstimulating	Limited Use, Accent Pieces
Yellow	Stimulating, Increases Alertness	Activity Areas, Playrooms
Pastels	Comforting, Relaxing	Ideal for General Living Spaces
Muted Shades	Neutral and Safe	Flexible Use Across Home

STRUCTURE

The product needed three layers: parent control, child

 **Subscription-aware flows**

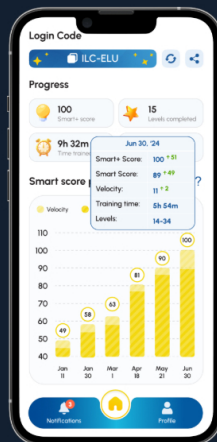
Each wireframe was crafted to ensure UX met user needs. Hidden complexity: subscription logic meant users with and without subscriptions experience different flows.



motivation, and training logic.

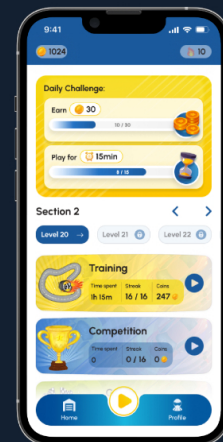
Three account types

We implemented three account types with different access levels to enable comprehensive parent control while keeping children engaged.



 Parent Dashboard

Track child's progress, Smart Score improvements, levels completed, and cognitive gains over time. Full control over settings and subscriptions.

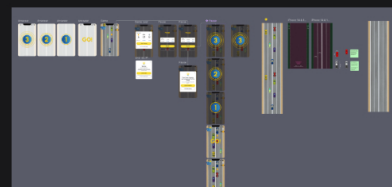


✦ Child Dashboard

Gamified experience with daily challenges, training sections, competitions, and fun rewards. Designed to feel like play, not work.

MOTIVATION

Children lose motivation easily, so the reward loop



became part of the learning system.

Through user surveys, we discovered this challenge and responded with a multi-layered motivation system.

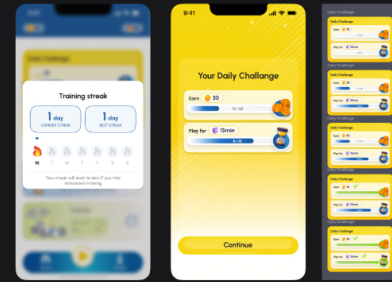
1 Complete two test levels → Unlock one mini-game play

2 Each level passed → Earn coins

3 Play the mini-game → Collect more coins

4 Spend coins → Play more mini-games

5 Complete daily challenges → Earn sparks



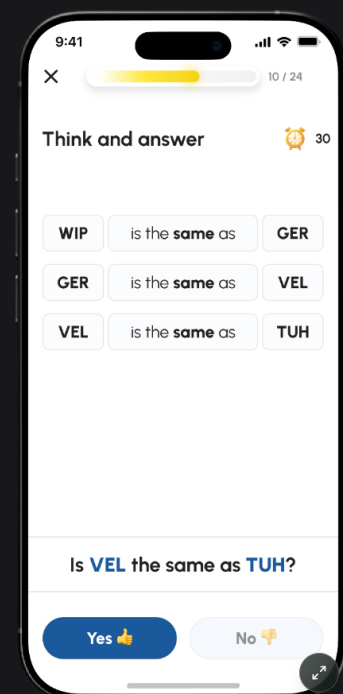
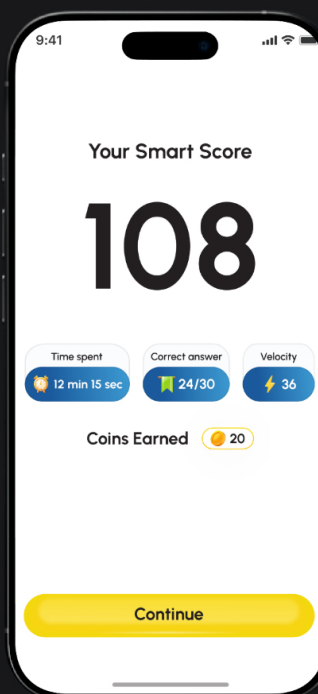
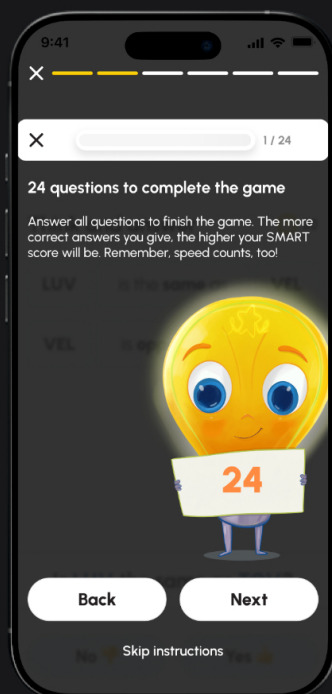
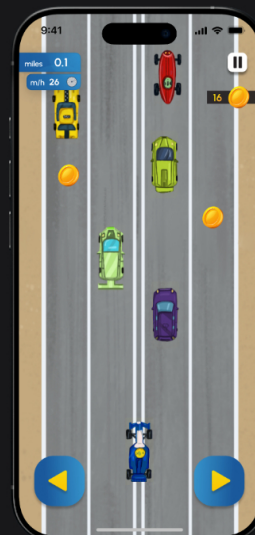
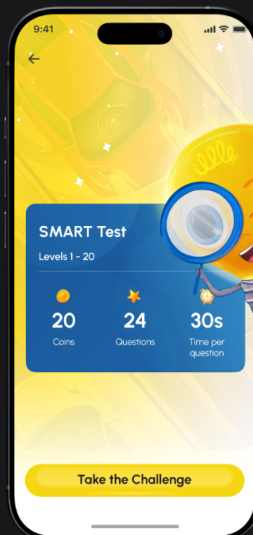
The Mini-Game Reward

A racing game where users collect coins. Tests give coins that can be used to play-creating a reward loop that keeps children engaged.



The IQ Test Experience

Gamified test start screen shows coins earned, questions count, and time per question. Designed to encourage children to take the challenge.



ACCESSIBILITY SYSTEM

Built for accessibility from the ground up, not retrofitted at the end.

Given the uniqueness of our UI requirements and accessibility needs, we built a comprehensive design system from scratch rather than using pre-made solutions.

Every element was designed with **WCAG 2.2 Level AAA compliance**—the highest accessibility standard achievable.



Intuitive Timers

Visual timers showing how long until screens change—helping children adapt



Sound Control

Mute or adjust sound effects, avoiding unexpected alarms or triggers



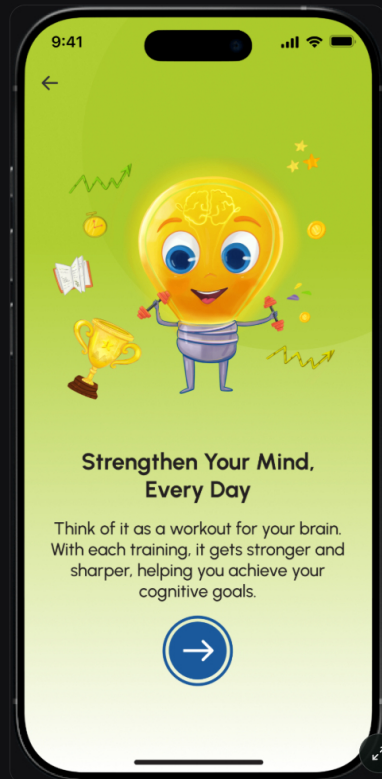
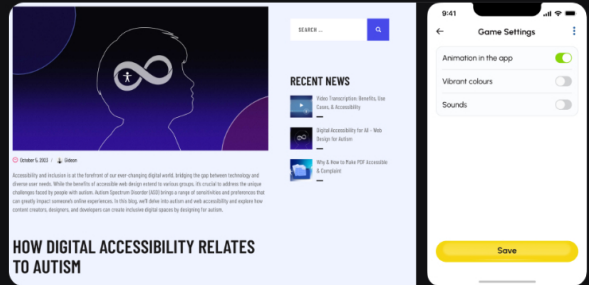
Animation Control

Pause, mute, or customize animations to prevent overstimulation



Color Adjustment

Brightness and color schemes adjustable per user



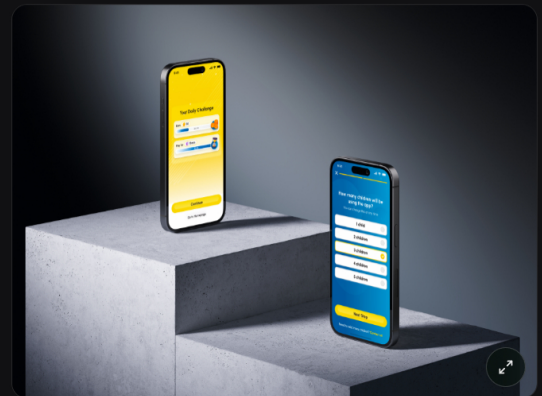
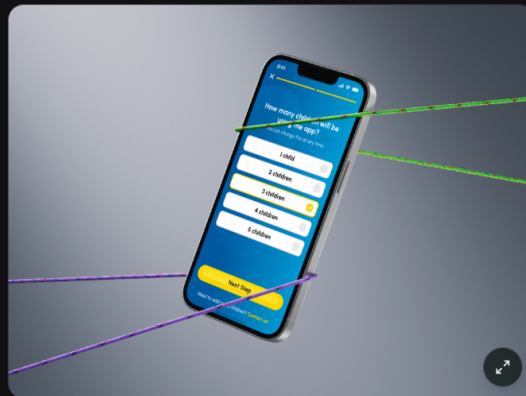
ONBOARDING

Maxot Glowie: the friendly guide.

Created through detailed technical specifications, Maxot became more than branding—he became the friendly guide who explains training rules during onboarding.

"Think of it as a workout for your brain. With each training, it gets stronger and sharper, helping you achieve your cognitive goals."

Where science meets design.



AAA

WCAG 2.2 Level AAA
Highest accessibility standard

4mo

Concept to Launch
Full design and development



"Dylan's social skills really improved. He has also improved academically. He didn't mix with other children before, but now he does."



"Math class is easier for me now and I am quicker at figuring things out in all my subjects. I feel like I am faster at everything in school now."

KEY TAKEAWAYS

What this project taught me

1

Research isn't optional

When your users experience the world differently than you do, design research becomes the foundation of your design.

taught me.

deep research becomes the foundation of every design decision.

2

Accessibility requires collaboration

Achieving the highest standards requires designers, developers, illustrators, scientists, and parents working together.

3

Customization is fundamental

Control over colors, sounds, and animations aren't features to add later—they're essential to inclusive design.

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dobzha

Senior Product Designer, mentor and AI Design Engineer
— building mobile apps and web platforms to make the internet more fun.

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